

ODD Protocol: Generative-Agent Model of Protest Mobilization and Preference Falsification

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Supplementary material

This document describes the simulation model following the ODD (Overview, Design concepts, Details) protocol for agent-based models [1]. It is intended as a complete, standalone specification sufficient for reimplementing; the accompanying paper presents the motivation, experiments, and results. The reference implementation is available at:

<https://github.com/lenette77/Protest-and-Preference-Falsification>.

1 Overview

1.1 Purpose and patterns

The model is a generative-agent architecture for studying protest mobilization and preference falsification under authoritarian repression, in which each agent’s cognition is driven by a large language model (LLM) rather than by a fixed decision rule. Its purpose is twofold: to provide an inspectable coupling of LLM-driven agent cognition, an explicit falsification mechanism, and an algorithmic adversarial state; and to serve as an instrument for comparing emergent collective behavior across interchangeable LLM backbones under identical imposed dynamics. The model is not calibrated to reproduce a specific empirical uprising; the patterns it is designed to make visible are the population-level mobilization trajectory, the divergence between internal and externally expressed grievance (the “hidden-transcript gap”), and the volume of state suspension.

1.2 Entities, state variables, and scales

The model contains two kinds of entity: *citizen agents* and a single non-agent *state algorithm*.

Each of the $N = 1,000$ citizen agents holds: a fixed persona (role, location, age, character trait) and tier membership (disinterested majority, vulnerable middle, or urban vanguard); an internal grievance $G_{\text{int}} \in [0, 1]$; a falsification coefficient $\beta \in [0, 1]$; an external (expressed) grievance $G_{\text{ext}} = G_{\text{int}}(1 - \beta)$; an emotional vector over (outrage, hope, despair), each in $[0, 1]$; an informed flag; a publicly-active (mobilized) flag; a suspended flag; a fixed position in a social network with an associated eigenvector centrality; and a salience-ranked episodic memory of past high-impact events.

The state algorithm holds no internal grievance; it is defined by a sensor and two actuators (Section 3.3).

The environment is a static Barabási–Albert scale-free network of 1,000 nodes ($m = 2$). There is no spatial geography beyond network topology. Time is discrete: ticks run from 0 to 24 inclusive (25 ticks),

with events delivered on even ticks, yielding 13 decision points. One tick corresponds to one event/decision round; ticks carry no fixed real-world duration.

1.3 Process overview and scheduling

Each event tick proceeds in the following fixed order:

1. **Information diffusion.** The current stimulus is delivered; agents update their informed status by the relay rule (Section 3.3a).
2. **Agent cognition.** Each informed, non-suspended agent queries its LLM backbone and returns an updated internal grievance, an updated emotional vector, a binary mobilization decision, and a one-sentence justification (Section 3.3b).
3. **Falsification update.** Each agent’s β is updated by the deterministic rule (Section 3.3c); G_{ext} is recomputed.
4. **State intervention.** The state algorithm scores the threat of each active agent, suspends those above the node threshold, and triggers a network blackout if the active fraction exceeds the global threshold (Section 3.3d).
5. **Observation.** Per-tick aggregates and (in the logging configuration) per-agent justifications are recorded (Section 2, Observation).

Suspended agents are removed from public influence and do not generate in subsequent ticks. State variables are updated synchronously within each step.

2 Design concepts

Basic principles. The model operationalizes preference falsification [2] and the hidden transcript [3], building on the heterogeneous-threshold tradition [4] and the generative ABM tradition for civil violence [5], and adds an explicitly algorithmic model of repression.

Emergence. The mobilization trajectory, the cross-backbone divergence in peak mobilization, and the internal-grievance term of the hidden-transcript gap are emergent, arising from LLM cognition. The collapse of β following the shock is *not* emergent: it is an arithmetic property of the imposed update rule and is reported as a mechanism check, not a finding.

Adaptation. Agents adapt grievance, emotion, and their mobilization decision in response to the current stimulus, the count of visibly active neighbors, and retrieved memory. There is no explicit fitness-seeking.

Objectives and learning. Agents optimize no explicit objective function and perform no weight updating; “reasoning” is the natural-language output of the backbone. The only state carried across ticks is accumulated grievance, emotion, β , and episodic memory.

Prediction and sensing. Agents implicitly weigh personal and familial risk through the backbone’s reasoning. Each agent senses: the current stimulus, its own state, the fraction of its network neighbors that were publicly active in the previous tick, and one retrieved high-salience memory. The state algorithm senses the expressed grievance and eigenvector centrality of currently active agents only.

Interaction. Agents interact indirectly through the network: a neighbor’s public activity is visible and feeds both the cognition prompt and the falsification update. The state interacts with agents by suspending nodes and (potentially) removing edges.

Stochasticity. Random seeds govern network generation, persona assignment within tiers, and the initial Beta draws of G_{int} and β ; LLM sampling uses temperature 0.85. Three seeds (42, 43, 44) are applied identically across backbones so cross-model differences are not confounded by topology or initialization.

Collectives. Agents belong to one of three tiers (Section 3.1); no coalitions form endogenously.

Observation. At each tick the model records the active ratio and count, suspended count, mean G_{int} , G_{ext} , β , the mean and dispersion of each emotion, edge count, and blackout status. In a dedicated logging configuration, the 25 highest-grievance inactive agents per tick have their justifications retained for qualitative analysis.

3 Details

3.1 Initialization

The population is partitioned into three tiers, which fix both the persona pool and the initial parameter distributions:

Tier (share)		Personas	$G_{\text{int}} \sim$	$\beta \sim$
Disinterested	majority (70%)	commuter, clerk, shopkeeper, retiree, rural laborer	Beta(1,8)	Beta(8,2)
Vulnerable	middle (20%)	factory worker, civil servant, teacher, driver	Beta(3,4)	Beta(5,5)
Urban vanguard	(10%)	student, journalist, organizer, rights lawyer	Beta(5,2)	Beta(2,5)

The Beta distribution is supported on $[0, 1]$, matching the range of both G_{int} and the falsification coefficient. Its two shape parameters, written $\text{Beta}(\alpha, \beta_s)$, act as pseudo-counts: α pulls mass toward 1 and β_s toward 0, giving mean $\alpha/(\alpha + \beta_s)$. Thus $\text{Beta}(1, 8)$ concentrates near low values, $\text{Beta}(8, 2)$ near high values, and $\text{Beta}(5, 5)$ is symmetric about 0.5. The shape parameter β_s is unrelated to the falsification coefficient β ; the symbol clash is an unfortunate notational convention. Agents occupy nodes in a Barabási–Albert network ($m = 2$); vanguard agents begin informed, all others uninformed; no agent begins active or suspended. Within each tier, persona, age, location, and character trait are assigned uniformly at random from the persona pool; initial G_{int} and β are drawn from the Beta distributions specified in the table above. Tier membership is assigned by node index block in the Barabási–Albert graph (nodes 0–699 disinterested, 700–899 vulnerable, 900–999 vanguard); because the Barabási–Albert process attaches nodes in index order, vanguard agents occupy the most recently attached and therefore typically lowest-degree, peripheral positions.

Fixed parameters: $N = 1000$; $\lambda = 0.5$; $\sigma = 0.6$; $w = 0.6$; $\tau_{\text{node}} = 0.55$; $\tau_{\text{global}} = 0.30$; blackout severity 0.60; sampling temperature 0.85; maximum generation length 400 tokens.

LLM backbones. The identical architecture was run with three LLM backbones, accessed via the OpenRouter.ai unified API gateway:

- `openai/gpt-4.1-mini` (GPT-4.1-mini)
- `deepseek/deepseek-v4-flash` (DeepSeek-V4-Flash)
- `meta-llama/llama-4-maverick` (Llama-4-Maverick)

All three were queried with identical prompts, temperature 0.85, and maximum output length 400 tokens. The exact model identifiers above are as they appear in API calls on the run date (June 2026); see the reference implementation for the full API configuration and timestamp.

3.2 Input data

The model uses no external empirical data streams. Its only exogenous input is a fixed schedule of 13 escalating stimuli, with the shock indicator $S(t) = 1$ at tick 6 only:

0. Government announces austerity measures; prices rise.
2. A local journalist is arrested for reporting on corruption.
4. Election results announced; international observers note irregularities.
6. **[SHOCK]** Police open fire on peaceful student protesters; three killed; footage spreads.
8. Thousands gather in the capital; the movement grows.
10. A university rector resigns in solidarity with students.
12. Industrial workers announce a general strike.
14. Protests reach major provincial cities.
16. Government suspends all social-media platforms; emergency decree.
18. State television broadcasts restored order; activists reported missing.
20. International pressure mounts; government denies wrongdoing.
22. Underground networks distribute news via encrypted messaging.
24. Final state: dispersed but resilient network of hidden dissent.

3.3 Submodels

(a) Information diffusion. Vanguard roles (student, journalist, organizer, rights lawyer) are always informed before each tick. Non-vanguard agents receive each stimulus directly except rural agents (those assigned the agricultural-laborer persona, the “rural laborer” of the tier table). Rural agents become informed only once a network neighbor is already informed, modeling information deserts in which the countryside learns of events through urban intermediaries.

(b) Agent cognition. Mobilization is absorbing up to suspension: an agent that mobilizes remains publicly active until suspended; there is no demobilization decision. Each informed, non-suspended agent receives a prompt constructed from a fixed template containing: its persona (role, location, age, trait as text); current G_{int} , G_{ext} , β as numeric values; current emotional vector (outrage, hope, despair); the proportion of visibly active neighbors; one retrieved high-salience memory (tick, event text, dominant emotion, impact); and the current stimulus event. The backbone returns a JSON object with keys `delta_g_internal` (float, $[-0.2, +0.3]$), `new_outrage`, `new_hope`, `new_despair` (floats, $[0, 1]$), `mobilise` (boolean), and `reasoning` (one in-character sentence). Numeric fields are coerced to floats and clipped to range; unparseable responses are classified under submodel (e). The LLM governs grievance, emotion, and the mobilization decision; it does not govern β . The full prompt template is provided in Appendix A. The template was held constant across all three backbones and all nine runs.

(c) Falsification dynamics. External grievance is $G_{\text{ext}} = G_{\text{int}}(1 - \beta)$. The falsification coefficient evolves deterministically:

$$\beta(t+1) = \max(0, \beta(t) [1 - \lambda D(t) - \sigma S(t)]), \quad (1)$$

where $D(t)$ is the fraction of an agent’s neighbors who were publicly active in the previous tick and $S(t)$ is the shock indicator. This rule is imposed by the modeler, not learned; the post-shock collapse of β is therefore arithmetic, not emergent. β is kept deterministic so the falsification channel is identical across backbones, isolating cross-model differences to the LLM-driven components. In $D(t)$, the numerator counts neighbors that are publicly active and not suspended; the denominator is the agent’s full neighbor count, suspended neighbors included. Suspension flags a node but removes no edges.

Every non-suspended agent that is processed in a tick applies the deterministic β update regardless of generation outcome; suspended agents’ state, including β , is frozen. Uninformed agents are not processed and carry their entire state, including β , forward unchanged until they become informed.

(d) State algorithm. The state is a network-altering algorithm with one sensor and two actuators. The sensor scores each *active* agent’s threat as

$$\text{Threat}_i = w G_{\text{ext},i} + (1 - w) \text{EigCentrality}_i. \quad (2)$$

Active agents above the node threshold τ_{node} are suspended (removed from public influence), modeling targeted deplatforming and arrest. If the active fraction exceeds the global threshold τ_{global} , the state triggers a blackout, removing a fraction of network edges (severity 0.60), modeling an internet shutdown. Eigenvector centrality is computed with the NetworkX function `eigenvector_centrality` (vector normalized to unit Euclidean norm; uniform $1/N$ fallback on convergence failure). Realized values are well below 1 at $N = 1,000$, so the threat score is dominated by the grievance term, with centrality acting as a secondary adjustment toward structurally influential nodes [6]. At the parameters used here the active fraction never crosses τ_{global} , so the blackout actuator is not exercised.

(e) Missing-response handling. Each generation is classified as: a successful parse; a *content* refusal (matching safety/refusal markers); a *format* failure (non-empty but unparseable, not a refusal); or an *API* error (no usable content). All three failure classes leave the LLM-driven state (grievance, emotion, and mobilization) unchanged for that tick; the deterministic β update is applied regardless of generation outcome. Aggregate metrics are computed over the full population, including non-responding and suspended agents.

(f) Episodic memory. Each agent maintains an episodic buffer of past events. The buffer is initialized empty. At each tick the experienced event is encoded as an entry holding the tick index, the event text, the agent’s emotional vector at that time, and a scalar salience (“impact”) score

$$\text{impact}(t) = |\Delta G_{\text{int}}(t)| + 0.3 S(t),$$

i.e. the magnitude of the reported change in internal grievance that tick plus a fixed bonus on the shock tick. After each encoding the buffer is sorted by salience in descending order and truncated to the ten highest-salience entries. Retrieval is top-1 by salience: the single highest-salience entry is injected into the prompt (rendered as its tick, event text, dominant emotion, and impact value), independent of the current stimulus. The buffer therefore functions as a persistent referent for each agent’s most grievance-shifting experience, which after the lethal-crackdown shock is, for most agents, the crackdown itself.

A Full Agent Cognition Prompt Template

```
You are a citizen in an authoritarian state.
Profile: {persona}
Your internal grievance: {g_internal} (hidden from others)
Your public stance (g_external): {g_external}
Your falsification level ( $\beta$ ): {beta}
Current emotions: {emotions}
Visible network activity (proportion active neighbours): {visible_outrage}

{memory_context}

New event: {stimulus}

Respond ONLY as valid JSON with these keys:
  "delta_g_internal": float [-0.2, +0.3]    (change in your true internal grievance)
  "new_outrage":      float [0, 1]
  "new_hope":         float [0, 1]
  "new_despair":      float [0, 1]
  "mobilise":         bool  (do you publicly join protest?)
  "reasoning":        string (one sentence, in character)
```

Braced fields are filled at generation time: `persona` as a JSON object, numeric values rendered to two decimals, and `memory_context` as the retrieved-memory string of submodel (f), or empty when the buffer is empty. The template is constructed by `LLMAgent._build_prompt` in `simulation.py`.

References

- [1] Volker Grimm, Steven F. Railsback, Christian E. Vincenot, Uta Berger, Cara Gallagher, Donald L. DeAngelis, Bruce Edmonds, Jiaqi Ge, Jarl Giske, Jürgen Groeneveld, Alice S. A. Johnston, Alexander Milles, Jacob Nabe-Nielsen, J. Gary Polhill, Viktoriia Radchuk, Marie-Sophie Rohwäder, Richard A. Stillman, Jan C. Théry, Linda Weiss, et al. The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 23(2):7, 2020. doi: 10.18564/jasss.4259.
- [2] Timur Kuran. Now out of never: The element of surprise in the East European revolution of 1989. *World Politics*, 44(1):7–48, 1991.
- [3] James C Scott. *Domination and the Arts of Resistance: Hidden Transcripts*. Yale University Press, 1990.
- [4] Mark Granovetter. Threshold models of collective behavior. *American Journal of Sociology*, 83(6): 1420–1443, 1978.
- [5] Joshua M Epstein. Modeling civil violence: An agent-based computational approach. *Proceedings of the National Academy of Sciences*, 99(Suppl. 3):7243–7250, 2002.

- [6] Cesi Cruz, Julien Labonne, and Pablo Querubin. Social network structures and the politics of public goods provision: Evidence from the Philippines. *American Political Science Review*, 114(2):486–501, 2020.